

ArgiSense Firmware Overview

This repository contains two product-specific firmware profiles. Keep their architecture notes separate because the boards have different sensor front-ends, pin maps, measurement flows, calibration data, and DAC output policies.

Product Firmware Documents

Product	Board target	Firmware source	Overview
Methane + pressure	argsense_mp_u575rg	app/products/methane_pressure/src	docs/methane-pressure-firmware-overview.md
pH	argsense_ph_u575rg	app/products/ph/src	docs/ph-firmware-overview.md

Shared Documents

Document	Purpose
docs/sensor-io-mapping.md	Consolidated IO mapping for both boards, including schematic conflicts and open hardware items.
docs/dynament-platinum-methane.md	Dynament Platinum methane sensor protocol and electrical notes.
docs/htu21d-humidity.md	HTU21D humidity and ambient temperature notes for the methane + pressure product.
docs/sensor-platform-v1-ph-analysis.md	pH schematic analysis and pH front-end implementation notes.

Build Targets

Methane + pressure:

```
py -3.12 -m west build -p always -b argsense_mp_u575rg argsense-zephyr-app
```

pH:

```
py -3.12 -m west build -p always -b argsense_ph_u575rg argsense-zephyr-app
```

Both product profiles share the common services under `app/common/src`, but sensor measurement logic and product policy live under the matching `app/products/<product>/src` directory.